

Estimation: the parameters, and how each is identified

Parent block

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Purpose

This records the specification you gave, what identifies each parameter, and one question where two parameters turn out to compete for the same variation. The model equations are reproduced first so the identification arguments can be read against them.

Everything below is implemented and the model solves.

1. The model

Preferences

The parents' period utility, with $l_{p,t} = 1 - h_{p,t} - \tau_{p,t}$:

$$U_{p,t} = \phi_1 \frac{c_{p,t}^{1-\rho}}{1-\rho} + \phi_2 \frac{l_{p,t}^{1-\eta}}{1-\eta} + \phi_3 \ln HC_t$$

Parental leisure nets out **both** work and time with the child, so time with the child is priced at the same rate as time working. The child's period utility, with $l_{c,t} = 1 - \tau_{p,t} - i_{c,t}$:

$$U_{c,t} = \lambda_1 \ln l_{c,t} + \lambda_2 \ln HC_t$$

The family's joint utility, with the parents' welfare weight $\tilde{\mu}_t$:

$$U_{f,t} = \phi_1 \frac{C_t^{1-\rho}}{1-\rho} + \phi_2 \frac{l_{p,t}^{1-\eta}}{1-\eta} + \tilde{\alpha}_1 \ln l_{c,t} + \tilde{\alpha}_2 \ln HC_t$$

$$\tilde{\alpha}_1 = (1 - \tilde{\mu}_t) \lambda_1, \quad \tilde{\alpha}_2 = \tilde{\mu}_t \phi_3 + (1 - \tilde{\mu}_t) \lambda_2$$

So the weight on the child's leisure is set entirely by the child's welfare share, while the weight on skill mixes the parents' valuation ϕ_3 and the child's λ_2 .

The welfare weight is $\tilde{\mu}_t = 1$ for $t < 6$ — during childhood the parents decide alone — and declines thereafter:

$$\tilde{\mu}_t = \mu_0 + \mu_1 (t - 5), \quad t \geq 6$$

Human capital

$$HC_{t+1} = R_t \tau_{p,t}^{\sigma_{1,t}} e_{p,t}^{\sigma_{2,t}} HC_t^{\sigma_{3,t}} i_{c,t}^{\sigma_{4,t}}, \quad \sigma_{j,t} = \exp(\sigma_{j0} + \sigma_{j1}(t - 1))$$

τ_p is parental time, e_p monetary investment, i_c the child's own study time, which enters only from $t = 6$. Self-productivity $\sigma_3 < 1$ is required for a bounded solution.

Budget

$$a_{t+1} = (1 + r)a_t + y + \lambda(w_t h_{p,t})^{1-\tau} - c_{p,t} - e_{p,t}, \quad a_{t+1} \geq 0$$

with the HSV/Benabou progressive tax; τ is progressivity and λ the level.

Terminal value at separation

$$V^{\text{term}} = \psi \ln k + \kappa_{\text{term}} \ln a$$

where k is the child's human capital at 18 and a the parents' retained assets.

2. What changed

The preference weights are time-invariant. $\phi_1, \phi_2, \phi_3, \lambda_1, \lambda_2$ were age profiles $\phi_{j0} + \phi_{j1}(t - 1)$; they are now scalars.

$\phi_1 = 1$ **and** $\lambda_1 = 1$. Utility is defined only up to *relative* weights, so two of the five must be normalised. ϕ_1 is fixed at 1 as you specified; λ_1 is fixed at 1 for the same reason on the child's side.

$\psi = 0$. The parents no longer place a separate terminal weight on the child's skill; they value it through the flow term $\phi_3 \ln HC_t$ and through altruism.

ψ had previously been raised from 1.0 to 4.0 because a low terminal weight made the final period value skill far less than every earlier one, and τ_p collapsed at $t = 17$. **I checked whether that recurs at zero. It does not** — the model solves and τ_p runs $0.570 \rightarrow 0.239$ at $t = 13 \rightarrow 0.314$ at $t = 17$. The flow term and altruism carry what the terminal weight used to.

Initial human capital comes from the data. HC_0 was an arbitrary $U(0, 1)$ draw. The model's human capital is now measured in the units of the Woodcock–Johnson PCA composite, and HC_0 is drawn from the data: the composite is not administered before age 3, so the mean and standard deviation of $\ln HC$ are fitted log-linearly on ages 3–17 and extrapolated back to age 0, giving $\ln HC_0 \sim N(5.929, 0.070)$ — a median of 376 and a 10–90 range of 344–411. The simulated draw reproduces this (median 375.6, p10 344, p90 412).

3. The parameters and what identifies each

Parameter	Role	Identified by
ϕ_1	weight on consumption	fixed at 1 (normalisation)
ϕ_2	weight on parental leisure	mean hours worked
ϕ_3	parents' weight on the child's skill	parental time and monetary investment
λ_1	child's weight on own leisure	fixed at 1 (normalisation)
λ_2	child's weight on own skill	the child's own study time
μ_1	slope of the welfare weight	see the question in §5

ϕ_2 works through the intratemporal condition: it prices leisure against the after-tax wage, so mean hours pin it.

ϕ_3 and λ_2 are the two valuation parameters. ϕ_3 raises both forms of parental investment — time and money — because both enter the same production function; λ_2 raises the child's own study time, which is the margin the child controls once $\tilde{\mu}_t < 1$.

One consequence worth stating. Valuation and technology both raise investment: a higher ϕ_3 and a more productive technology look the same in τ_p and e_p alone. What separates them is the *resulting level of human capital* — technology raises it, valuation does not. This is why putting *HC* into the data's units mattered: it makes the skill level usable as a moment, and without it ϕ_3 and λ_2 cannot be told apart from the production-function elasticities.

4. For the next step

These are not estimated yet. Listed with what would identify each.

Parameter	Role	Would be identified by
κ_{term}	weight on the parents' retained assets in their terminal value	parental net worth after the child leaves — family assets at child age 17, and post-separation wealth for families whose child has finished college
κ_0	level of the psychic cost of college	the overall college enrolment rate
κ_θ	ability gradient in the psychic cost (negative)	the enrolment gradient in child test scores — enrolment by achievement quantile
κ_{ParEd}	parental-education shift (negative)	the enrolment gap by parental education

The three κ terms of the psychic cost all move enrolment, so they need three *distinct* enrolment moments: the level, the slope in ability, and the gap by parental education. Estimated against the overall rate alone they would not be separately identified.

κ_{term} is a different object: it governs how much wealth parents retain rather than transfer, so post-separation wealth identifies it, not enrolment.

Their present values are transported from Colas (Table 2) by **sign and ratio only** — $\kappa_{\text{ParEd}}/\kappa_{\theta} = 0.205$ is hers. The levels are not transportable: her utility is $c^{1-\gamma}/(1-\gamma)$ at $\gamma = 1.9$ with consumption in dollars, ours is $\rho = 1.5$ with c of order 0.1–5.

5. Question

Can μ_1 and σ_{41} both be estimated?

Your suggestion was that μ_1 is identified from ϕ_3 and λ_2 — only ϕ_3 operates before age 6, all three after. That is right in direction, and with $\lambda_1 = 1$ it sharpens: the child’s effective weight on its own leisure is exactly $(1 - \tilde{\mu}_t)$, so λ_2 **sets the level of the child’s study time** and μ_1 **sets its slope in age**.

The difficulty is that the child’s first-order condition for study time is driven by

$$\frac{\sigma_{4,t}}{(1 - \tilde{\mu}_t) \lambda_1}, \quad \sigma_{4,t} = \exp(\sigma_{40} + \sigma_{41}(t - 1))$$

and σ_{41} and μ_1 both bend that ratio with age. **The age profile of study time therefore cannot identify both** — they enter as a single combination, and an optimiser given both would sit on a ridge and return whichever point it started nearest.

One of the two must be fixed. Provisionally I have held σ_4 flat in age ($\sigma_{41} = 0$) and left μ_1 at its current value, on the reasoning that μ_1 is a welfare weight fixed by the structure of the problem while σ_4 is a technology parameter that can reasonably be constant. But the choice is yours, and there are three coherent options:

1. **Fix μ_1 by an explicit rule and estimate σ_{41} .** For instance, set μ_1 so the parents’ weight reaches 0.5 at $t = 17$ (equal weights as the child leaves), giving $\mu_1 = -1/24$; or so it reaches 0 at $t = 17$ (the child fully decides at separation), giving $\mu_1 = -1/12$.
2. **Estimate μ_1 and hold σ_4 flat.** This is what is currently implemented. It treats the age profile of study time as evidence about the child’s growing autonomy rather than about the technology.
3. **Fix both** from outside the estimation, and use the study-time profile as an untargeted check on whether the model reproduces it.

If you have a rule in mind for μ_1 , option 1 is the one that buys the most: it makes the technology’s age profile an estimated object rather than an assumption.